Lesson Notes

MODULE 6 | LESSON 1

FACTOR INVESTING: PROFITABLE ANOMALIES OR ANOMALOUS PROFITS

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the previous module, we reviewed some of the most important portfolio optimization strategies and their relation to modern portfolio theory and the mean-variance optimization framework and then learned about extensions of these strategies based on more complex models or modified assumptions. For the most part, however, those strategies were more or less grounded in the capital asset pricing model (CAPM), which only allows a single "market factor" to determine the characteristics of an optimal portfolio. In this module, we go both forward, beyond the CAPM, and backward, revisiting the factor model introduced in Module 2 of this course. Factor investing incorporates several interesting techniques to identify and validate the relevant factors, many of which we learn in this lesson.

1. What We Already Know About Factor Models

In Module 2, Lesson 4, you learned a lot about factor models, including but not limited to the following:

- The different kinds of factors:
  - **Macroeconomic factors:** observable economic or financial time series, like inflation or GNP growth
  - **Fundamental factors:** created from observable asset characteristics, including accounting ratios like price-to-book value (P/BV) and earnings per share (EPS)
  - **statistical or latent factors:** not observable or extracted from directly from asset returns, like principal components
- The assumptions underlying factor analysis:
  - The factors must be stationary and their unconditional moments are given by:

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$$E[f_t] = \mu_f, \quad \text{cov}(f_t) = \Sigma_f.$$

- The error terms are uncorrelated across factors.
  - The error terms are serially uncorrelated but contemporaneously correlated across assets.
- The relationships between covariance, coskewness, and cokurtosis of factors, e.g., for a factor model

$$R_t = a + Bf_t + \epsilon_t,$$

where B is the matrix of factor loadings and F the matrix of factors, then the following is true:

$$M_3 = BM_{3,f}(B^T \otimes B^T) + \Omega$$
$$M_4 = BM_{4,f}(B^T \otimes B^T \otimes B^T) + \Gamma.$$

where $M_{3,f}$ and $M_{4,f}$ are the coskweness and cokurtosis matrices for the factors, and Ω and Γ are special matrices (described in more detail in Module 2, Lesson 4).

- The Fama-French three-factor model below (as well as some of its variations):

$$R_{it} - r_{ft} = \alpha_{it} + \beta_1(R_{Mt} - r_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \epsilon_{it}$$

where:

R_{it} is the return of asset i at time t;

r_{ft} is the risk-free rate of return at time t;

R_{Mt} is the return on the market portfolio;

$R_{it} - r_{ft}$ and $R_{Mt} - r_{ft}$ are excess returns on asset i and the market portfolio, respectively;

$R_{Mt} - r_{ft}$ is the market premium (momentum factor);

SMB_t is the size premium (small minus big).

HML_t is the value premium (high minus low).

2. Beyond CAPM: Arbitrage Pricing Theory

In this lesson, the main reading briefly reviews some of this theory (both at the beginning and the end of the reading), with an important reference to arbitrage pricing theory (APT): The factors of APT would be considered mere anomalies by the capital asset pricing model (CAPM). Assuming such factors reliable exist, APT can be seen as a corrective update to the inadequate CAPM.

From there, we start sifting through the practical issues of factor investing, from the very beginning —the methodology for factor identification and validation—to the very end: recognizing the slow death of the anomaly known as alpha decay. We walk through the portfolio sorting process, which allows us to verify and visualize the existence of factors, taking as our example the best known ones, including size, value, and momentum. Abasi et al. provide a nice review, as well as Python code implementation of APT.

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3. What Are the Odds?

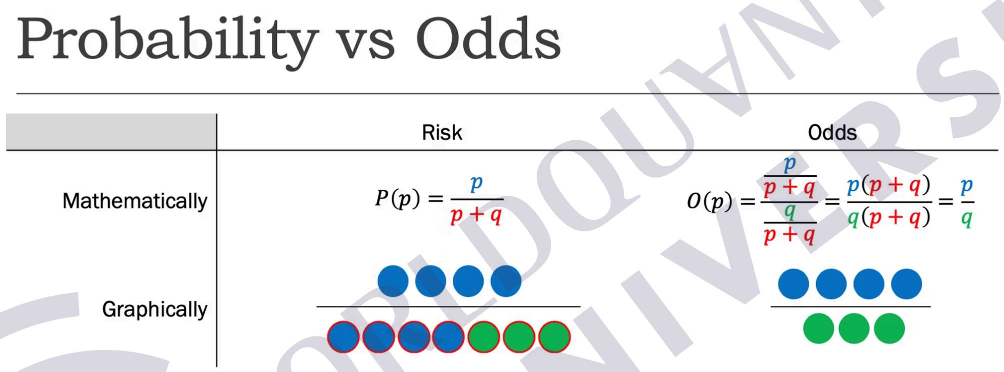
To establish the statistical significance of newly identified factors, we are encouraged to go beyond the standard regression p-values, in favor of Bayesianized p-values involving a user-specified prior. You will see that these Bayesianized p-values have the benefit of a more intuitive interpretation if you remember that "odds" can be defined as the ratio of the probability that an event will occur over the probability that an event will not occur.

For example, a fair coin has a 50% probability of landing on heads and a 50% chance of landing on tails, so the odds are 1 (=.5/.5).

The probability of a a die landing on 3 is 1/6. The probability of a a die *not* landing on 3 is 5/6. So the odds of the die landing on 3 are 1:5 or 1/5 =(1/6) / (5/6). The odds of the die *not* landing on 3 are 5:1 or 5.

The following figure is provided as a visualization of the concept of odds in contrast to "risk" or chance or probability.

Figure 1: Probability vs. Odds Visualization



Source: George, Andrew et al. "What's the Risk: Differentiating Risk Ratios, Odds Ratios, and Hazard Ratios?" *Cureus*. https://www.cureus.com/articles/39455-whats-the-risk-differentiating-risk-ratios-odds-ratios-and-hazard-ratios.

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4. The Fama-Macbeth Regression

We are then introduced to the Fama-Macbeth regression, which is actually a pair of regressions:

1.) We regress each asset's returns against each factor return stream (Abasi et al.):

$$R_{1,t} = \alpha_1 + \beta_{1,F_1} F_{1,t} + \beta_{1,F_2} F_{2,t} + \cdots + \beta_{1,F_m} F_{m,t} + \epsilon_{1,t}$$
$$R_{2,t} = \alpha_2 + \beta_{2,F_1} F_{1,t} + \beta_{2,F_2} F_{2,t} + \cdots + \beta_{2,F_m} F_{m,t} + \epsilon_{2,t}$$
$$\vdots$$
$$R_{n,t} = \alpha_n + \beta_{n,F_1} F_{1,t} + \beta_{n,F_2} F_{2,t} + \cdots + \beta_{n,F_m} F_{m,t} + \epsilon_{n,t}$$

2) We calculate the risk premia by using the beta estimates from the first step as our exogenous variables to estimate the mean return of each asset (Abasi et al.):

$$E(R_i) = \gamma_0 + \gamma_1 \hat{\beta}_{i,F_1} + \gamma_2 \hat{\beta}_{i,F_2} + \cdots + \gamma_m \hat{\beta}_{i,F_m} + \epsilon_i$$

While the first reading provides a nice overview of this technique (along with another Python implementation), the second reading provides some helpful details and additional elaboration (along with yet another Python implementation).

Please take note of the discussion of publication bias. The general upshot of this phenomenon is that *all* studies should be read and evaluated critically. We should always keep in mind that academics and other researchers do not make headlines with studies that show non-results. The importance of such a critical mindset will become even more clear in the next and future lessons.

5. Conclusion

In this lesson, we discussed the techniques and tests for identifying and validating factors. We also walked through the Python code that implements these techniques and tests. In the next lesson, we contextualize factor investing with a bit of history and several models to explain what went wrong before--and what could possibly go wrong again.

References

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- Coqueret, Guillaume, and Tony Guida. "Chapter 3: Factor Investing and Asset Pricing Anomalies." *Machine Learning for Factor Investing*. http://www.mlfactor.com/chap_3.html.
- George, Andrew et al. "What's the Risk: Differentiating Risk Ratios, Odds Ratios, and Hazard Ratios?" *Cureus*. https://www.cureus.com/articles/39455-whats-the-risk-differentiating-risk-ratios-odds-ratios-and-hazard-ratios.
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